CSA17 Artificial Intelligence: MCQ Test (CO1 and CO2)

Slot D (CSA1711) | 20 questions | Maximum marks: 20 | Time: 30 minutes

SET NUMBER
SET 09

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Signature	B.A. Raj	Date	31/10/2026

Instructions to students (read before you start)

1. Write your answers **only in the ANSWER BOX below**. Marks written next to the questions are **not counted**.
2. In each box write **ONE CAPITAL letter: A, B, C or D**, large and clear, with a **blue or black pen**. No pencil.
3. **No overwriting or corrections.** A box with two letters, an overwritten letter, a tick, or a blank box gets **0**.
4. Each question carries 1 mark. **No negative marks.** Do not write anything else inside the answer box.
5. Check that your Reg No and **SET number** are correct. Do not exchange papers. Return **both pages**.

ANSWER BOX: SET 09 — **Write A, B, C or D in each box**

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
A	B	B	D	C	A	C	B	C	A
Q11	**Q12**	**Q13**	**Q14**	**Q15**	**Q16**	**Q17**	**Q18**	**Q19**	**Q20**
A	B	A	C	B	A	B	B	B	C

For evaluator only: Marks obtained _____ / 20

Evaluated by Claude using the SET 09 key

PART A: CO1 (Unit I) | Questions 1 to 10

1. IDS with b = 10 and d = 5 generates how many nodes?
   (A) 50
   (B) 111,110
   (C) 100,000
   (D) 123,450

2. Edges: S-A, S-B, A-C, A-D, B-E, B-G. Goal G, children in alphabetical order. DFS visit order:
   (A) S, A, B, G
   (B) S, A, B, C, D, E, G
   (C) S, B, G
   (D) S, A, C, D, B, E, G

3. Edges: S-A, S-B, A-C, A-D, B-E, B-G. Goal G, children in alphabetical order. BFS visit order:
   (A) S, B, G
   (B) S, A, C, D, B, E, G
   (C) S, A, B, C, D, E, G
   (D) S, B, E, G

4. Costs: S-A = 1, S-B = 5, A-G = 10, B-G = 3. Uniform-cost search returns:
   (A) S-B-G with cost 5
   (B) S-B-G with cost 8
   (C) S-A-G with cost 1
   (D) S-A-G with cost 11

5. Agent = ?
   (A) Model + Utility
   (B) State + Goal
   (C) Architecture + Program
   (D) Sensor + Actuator

6. For a medical diagnosis system, which one is an ACTUATOR?
   (A) Display of tests and treatments
   (B) Healthy patient
   (C) Hospital
   (D) Keyboard entry of symptoms

7. A full binary tree has levels 0 to 3 (root at level 0). Total nodes =
   (A) 8
   (B) 16
   (C) 7
   (D) 15

8. The agent function maps:
   (A) States to goals
   (B) Percept sequences to actions
   (C) Sensors to actuators
   (D) Actions to percepts

9. In PEAS, the letter 'A' stands for:
   (A) Actions
   (B) Algorithms
   (C) Actuators
   (D) Agents

10. For an automated taxi, which one is a SENSOR?
(A) Camera (B) Accelerator
(C) Steering (D) Horn

PART B: CO2 (Unit II) | Questions 11 to 20

11. In Sudoku, 'all values in a row are different' (Alldiff) is a:
(A) Global constraint (B) Soft constraint
(C) Unary constraint (D) Binary constraint only

12. MAX root with 3 MIN children. Leaves: (3, 12, 8), (2, 4, 6), (14, 5, 2). Minimax value of root =
(A) 2 (B) 12
(C) 14 (D) 3

13. A* finds a goal at depth 5 after generating 52 nodes. Effective branching factor b* is about:
(A) 10.4 (B) 5
(C) 1.92 (D) 52

14. Straight-line distance as a heuristic for road travel is:
(A) Always zero (B) Inadmissible
(C) Larger than road distance (D) Admissible

15. Heuristics obtained from a relaxed problem are:
(A) Admissible (B) Inconsistent
(C) Always overestimating (D) Never useful

16. Alpha-beta prunes a branch when:
(A) alpha ≥ beta (B) alpha = 0
(C) alpha < beta (D) beta = infinity

17. In alpha-beta pruning, alpha is:
(A) The number of leaves (B) The best value found so far for MAX
(C) The depth limit (D) The best value for MIN

18. If h(n) = 0 for every node, A* behaves like:
(A) Greedy best-first search (B) Depth-first search
(C) Hill climbing (D) Uniform-cost search

19. 8-puzzle: h1 = misplaced tiles, h2 = Manhattan distance. Which is true?
(A) h1 dominates h2 (B) h1 = h2 always
(C) h2 dominates h1 (D) Both are inadmissible

20. Time complexity of minimax (branching b, depth m):
(A) O(b + m) (B) O(m^b)
(C) O(b^m) (D) O(bm)

*** End of question paper: copy your answers into the ANSWER BOX on page 1 ***